INTELLIGENT CAMPUS ENERGY DEMAND FORECASTING USING MACHINE LEARNING

PROJECT REPORT

;

Submitted by

THATHIKA M (7376241CS442)

VARSHNIE S K (7376241CS451)

In partial fulfilment for the award of the degree of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING



BANNARI AMMAN INSTITUTE OF TECHNOLOGY
(An Autonomous Institution Affiliated to Anna University, Chennai)
SATHYAMANGALAM-638401

ANNA UNIVERSITY: CHENNAI 600 025

NOVEMBER 2026

ACKNOWLEDGEMENT

We would like to enunciate heartfelt thanks to our esteemed Chairman **Dr.S.V.Balasubramaniam** and the respected Principal **Dr. C. Palanisamy** for providing excellent facilities and support during the course of study in this institute.

We are grateful to the Head of the Department, **Dr. Sasikala D** , **Department of Computer Science and Engineering** for her valuable suggestions to carry out the project work successfully.

We wish to express our sincere thanks to Faculty guide **Mrs. MANOCHITRA A S, Assistant Professor, Department of Artificial Intelligence and Data Science** , for her constructive ideas, inspirations, encouragement, excellent guidance, and much needed technical support extended to complete our project work.

We would like to thank our friends, faculty and non-teaching staff who have directly and indirectly contributed to the success of this project.

THATHIKA M

(7376241CS442)

VARSHNIE S K

(7376241CS451)

ABSTRACT

Educational campuses consume electricity across classrooms, laboratories, libraries, hostels and administrative buildings. Demand changes with time of day, weekday, academic schedules, weather and building activity. This project proposes an intelligent campus energy demand forecasting system using machine learning to estimate future consumption and support energy planning. The planned workflow includes collecting suitable historical energy data, validating and preprocessing records, exploring consumption patterns, creating time-based features, and comparing a seasonal-naïve baseline with candidate machine learning models such as Linear Regression, Random Forest and gradient-boosting regression. Chronological train-validation-test splits are proposed to preserve temporal order and reduce data leakage. Models will be assessed using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and Mean Absolute Percentage Error (MAPE) where appropriate. The proposed system also includes anomaly identification, electricity-cost and carbon-emission estimation using documented factors, and an interactive dashboard. This report describes a proposed/in-progress system. The available project report does not supply a verified dataset manifest, final model metrics or measured campus outcomes; therefore, no campus-specific accuracy, energy savings or emission reductions are claimed. Keywords: campus energy, electricity demand forecasting, machine learning, time-series validation, anomaly detection, dashboard, sustainability.

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ACKNOWLEDGEMENT	iv
	ABSTRACT	v
	TABLE OF CONTENTS	4
	LIST OF FIGURES	5
	LIST OF TABLES	6
1	**INTRODUCTION**	7
2	**LITERATURE SURVEY**	11
3	**METHODOLOGY**	14
3.1	System Architecture	14
3.2	Dataset Selection and Preparation	15
3.3	Data Preprocessing	18
3.4	Exploratory Data Analysis	19
3.5	Feature Engineering	19
3.6	Forecasting Models	20
3.7	Chronological Validation	20
3.8	Evaluation Metrics	21
4	**PROPOSED SYSTEM MODULES AND DESIGN**	22
4.1	System Workflow	22
4.2	Module Description	25
4.3	Dashboard Views	26
4.4	Cost and Carbon Estimation	27
4.5	Tools and Technologies	27

5	**RESULTS AND DISCUSSION**	28
5.1	Prototype Workflow	28
5.2	Functional Validation Plan	29
5.3	Evaluation Results	29
5.4	Discussion and Limitations	30
6	**CONCLUSION AND FUTURE WORK**	30
7	**INDIVIDUAL CONTRIBUTION**	31
8	**ORIGINALITY SCORE**	33
9	**REFERENCES**	30
	APPENDIX A – ENERGY FORECASTING QUESTION SET	34
	APPENDIX B – USER GUIDE	34

LIST OF FIGURES

FIGURE NO.	FIGURE NAME	PAGE NO.
3.1	Overall system architecture	14
3.2	Data preparation pipeline	17
3.3	Forecast model development and evaluation	18
4.1	Proposed dashboard views	22
4.2	Chronological validation workflow	23
4.3	Originality report to be inserted	23

LIST OF TABLES

TABLE NO.	TABLE NAME	PAGE NO.
3.1	Suggested dataset fields and purpose	14
3.2	Candidate forecasting models	16
4.1	Proposed modules and responsibilities	25
4.2	Cost and carbon estimation inputs	26
5.1	Functional validation matrix	31
5.2	Evaluation result to be completed from actual runs	32
7.1	Individual contribution allocation	33

CHAPTER 1

INTRODUCTION

Educational institutions consume significant amounts of electricity across classrooms, laboratories, libraries, hostels, and administrative buildings to support academic activities and daily operations. Electricity is essential for lighting, computers, laboratory equipment, ventilation, air conditioning, communication systems, and other campus facilities. As educational institutions expand their infrastructure and adopt modern technologies, the demand for electricity continues to grow, making efficient energy management an important requirement for reducing operational expenses, improving resource utilization, and supporting environmental sustainability. Effective energy management requires a clear understanding of how electricity is consumed, when demand is highest, and how consumption may change in the future. However, campus electricity demand is not constant and varies according to several factors, including the time of day, day of the week, academic schedules, building activities, occupancy patterns, seasonal changes, and weather conditions. Classrooms may experience higher consumption during lecture hours, laboratories may require electricity for computers and specialized equipment, and hostels may experience increased demand during mornings and evenings. Libraries and administrative buildings may follow different operating schedules, resulting in variations in their electricity consumption patterns. These differences make campus energy management a complex task, particularly when historical consumption records are analyzed manually or used only for preparing electricity bills. Traditional energy management methods provide information about previous electricity usage but may not adequately support future demand prediction, early identification of unusual consumption, or systematic energy planning. Manual analysis can also make it difficult to recognize recurring patterns across large datasets and compare consumption over different periods. Therefore, an intelligent and data-driven approach is required to analyze historical electricity readings, identify meaningful consumption patterns, and estimate future energy requirements. Machine learning offers an effective approach by

identifying relationships between historical consumption and relevant time-based or contextual variables. Instead of relying entirely on manual calculations, machine learning models can learn from available observations and generate predictions that support informed energy management decisions. The proposed project, Intelligent Campus Energy Demand Forecasting Using Machine Learning, aims to develop a structured framework for forecasting campus electricity demand by integrating data collection, data preprocessing, exploratory data analysis, feature engineering, forecasting model development, performance evaluation, anomaly identification, sustainability estimation, and interactive visualization. The proposed system begins with collecting suitable historical electricity consumption data from available sources. The dataset is examined to understand its structure, column names, timestamps, measurement units, sampling frequency, and overall quality. Data preprocessing is performed to identify and handle missing values, duplicate records, inconsistent timestamps, invalid readings, and other data quality issues that may affect forecasting performance. Proper preprocessing is essential because incomplete or inconsistent observations can introduce errors into the analysis and reduce the reliability of the predictions. After preprocessing, Exploratory Data Analysis (EDA) is performed to understand the distribution of electricity consumption and identify variations across different time periods. Time-series plots, hourly consumption profiles, weekday comparisons, monthly summaries, and building-wise visualizations can be used to examine demand patterns, depending on the information available in the dataset. Correlation analysis may also be performed to explore relationships among numerical variables. These visualizations help users understand historical consumption behavior and identify patterns that may be useful for forecasting. Following exploratory analysis, feature engineering is used to transform the available data into meaningful inputs for machine learning models. Potential features include the hour of the day, day of the week, month, weekend or holiday indicators, academic-calendar information, and lagged electricity consumption values representing previous observations. Rolling statistical features may also be considered when sufficient historical data are available. These features help the models represent recurring demand patterns and relationships between past and future electricity usage. However, all lagged

and rolling features must be constructed using only information available at the prediction time to prevent data leakage. The forecasting component considers a seasonal-naïve method as a baseline and candidate machine learning algorithms such as Linear Regression, Random Forest Regression, and Gradient Boosting Regression. The seasonal-naïve method provides a simple reference by using consumption from a comparable previous period when the dataset's sampling frequency supports such a comparison. Linear Regression provides an interpretable approach for modeling relationships between input variables and electricity consumption, while Random Forest and Gradient Boosting can represent more complex nonlinear relationships. The performance of these approaches depends on the characteristics of the dataset, the availability of relevant features, and the selected forecast horizon. Therefore, the most suitable model must be selected based on measured performance rather than assuming that a more complex algorithm will always provide better predictions. Since electricity consumption is a time-dependent process, the project proposes chronological train-validation-test splitting to preserve the temporal order of observations. The earliest observations are used for training, a subsequent period is used for validation and model selection, and the latest period is reserved for final testing. This approach helps prevent future observations from influencing model development and provides a more realistic assessment of forecasting performance. The candidate models are evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE), where appropriate. MAE measures the average absolute difference between actual and predicted consumption, RMSE gives greater weight to larger prediction errors, and MAPE expresses prediction error as a percentage when the actual values make this measure meaningful. Because MAPE can become unstable when actual consumption values are zero or close to zero, MAE and RMSE are emphasized when percentage-based evaluation is unsuitable. Comparing the models on the same chronological test period provides a consistent basis for assessing their predictive performance and identifying an appropriate forecasting approach. In addition to demand forecasting, the proposed system includes an anomaly identification component to highlight electricity consumption readings that differ from expected usage patterns. Such

readings may occur because of unusual building activity, changes in occupancy, measurement errors, or other operational conditions. The anomaly component is intended to help users identify observations that require further investigation rather than automatically classify them as faults or energy wastage. The project also aims to estimate electricity costs and associated carbon emissions to provide additional information about the financial and environmental implications of energy consumption. Electricity costs can be estimated by multiplying the relevant energy consumption by the applicable electricity tariff, while estimated carbon emissions can be calculated using a documented emissions factor expressed in suitable units. The reliability of these estimates depends on the accuracy of the consumption data and the suitability of the selected tariff and emissions factor for the relevant region and reporting period. These calculations provide estimates and should not be interpreted as proof of actual cost savings or emission reductions. To make the analysis accessible, an interactive dashboard is proposed to present historical consumption trends, forecast values, model comparison results, evaluation metrics, anomaly indicators, and sustainability estimates in a clear and understandable format. The dashboard may include filters for date ranges and building categories when the dataset supports these features. Actual consumption and predicted values should be visually distinguished so that users can interpret the results correctly. By combining forecasting and visualization, the system aims to reduce the effort required to examine historical energy data and provide a clearer understanding of expected demand. The primary objective of this project is to establish a systematic, reproducible, and data-driven framework that transforms historical campus electricity readings into meaningful forecasts and decision-support information. The specific objectives include preparing suitable energy data, examining consumption patterns, developing relevant forecasting features, comparing baseline and machine learning models, evaluating predictions using appropriate metrics, identifying unusual readings, estimating electricity costs and carbon emissions, and presenting the results through an interactive dashboard. The intended scope includes classrooms, laboratories, hostels, libraries, and administrative buildings, subject to the availability of suitable historical energy readings. Additional information, such as building identifiers, weather conditions, occupancy

indicators, and academic schedules, may be incorporated when reliable data are available. Nevertheless, the effectiveness of the system depends on the quantity, quality, and consistency of the historical data, the sampling interval, the forecast horizon, the availability of contextual variables, and changes in campus operating conditions. Missing readings, irregular measurement intervals, unexpected changes in building usage, and limited historical observations may affect prediction quality. Furthermore, anomaly flags require appropriate interpretation, and cost and carbon estimates must be supported by documented input factors. The proposed system is intended to support energy planning and analysis rather than directly control campus electrical equipment, replace professional energy audits, or guarantee reductions in electricity consumption, operating costs, or carbon emissions. Its effectiveness must be established through actual implementation, functional testing, chronological model evaluation, and documented experimental results. The project also emphasizes transparent reporting of dataset characteristics, preprocessing decisions, feature definitions, model settings, validation periods, evaluation metrics, and limitations to improve reproducibility and make the results easier to interpret. By integrating historical energy analysis, machine learning-based forecasting, anomaly identification, sustainability estimation, and interactive visualization, the proposed system aims to provide an effective foundation for understanding campus electricity demand, anticipating future consumption patterns, and supporting informed decisions toward more efficient and sustainable campus energy management.

\

CHAPTER 2

LITERATURE SURVEY

The development of intelligent document lifecycle management systems has become increasingly important as individuals and organizations manage digital documents such as certificates, insurance policies, contracts, licenses, and other records with specific validity periods. Traditional document management methods often depend on manual data entry, spreadsheets, and calendar reminders, which can make it difficult to maintain accurate records and track expiration dates. Optical Character Recognition (OCR) provides a way to convert text from scanned documents and images into machine-readable information. Tesseract OCR is an open-source technology that supports text extraction from images and scanned documents [1]. Although OCR can reduce manual data entry, its performance may be affected by poor image quality, complex document layouts, and unclear text. Furthermore, extracting text alone does not always provide enough information to understand the document's purpose or identify the correct expiration date. Multimodal artificial intelligence offers a more advanced approach by analyzing document content and visual structure together. Google's Gemini API documentation describes capabilities for processing documents and extracting information from their content [2]. These capabilities can support document classification and metadata extraction, including identifying document types, relevant dates, and associated organizations. However, AI-generated information may be incomplete or incorrect, especially when documents contain multiple dates or unclear information. Therefore, automated extraction should be combined with validation rules and human review to improve reliability.

Email integration provides another important opportunity to automate document collection and processing. Many important documents are received as email attachments, and manually downloading, organizing, and recording each attachment can require considerable time and effort. The Gmail API provides functionality for retrieving email

messages and accessing their attachments, enabling applications to process documents directly from a user's mailbox [3]. Its history functionality also supports identifying mailbox changes after a previously recorded history identifier, which can help applications synchronize newly received messages without repeatedly processing the entire mailbox [4]. Secure authorization is essential when connecting an application to an email account. The OAuth 2.0 Authorization Framework describes how applications can obtain authorized access to protected resources [5], while Proof Key for Code Exchange (PKCE) provides additional protection for authorization code flows [6]. These technologies provide a foundation for implementing controlled access to Gmail and retrieving relevant attachments. By combining email integration with document processing, a system can reduce manual uploads and organize documents more efficiently. However, the application must handle synchronization failures, expired authorization credentials, and changes in mailbox history to maintain reliable operation.

Security and data protection are also essential considerations because document management systems may process confidential personal, financial, and legal information. Sensitive credentials must be protected against unauthorized access, and stored authentication tokens should not be exposed as readable text. NIST Special Publication 800-38D describes Galois/Counter Mode (GCM), an authenticated encryption method that provides data confidentiality and integrity protection [7]. This provides a technical foundation for securely storing sensitive values, such as OAuth refresh tokens, when implemented with appropriate key management and secure storage practices. Document duplication is another issue that can arise when identical attachments are received multiple times or uploaded repeatedly. Cryptographic hashing, such as SHA-256, can help identify files with identical content and reduce unnecessary processing. However, different files may contain visually identical information, so content hashing alone cannot identify every possible duplicate. The OAuth 2.0 Security Best Current Practice document also provides guidance on reducing authorization-related security risks and protecting access credentials [8]. Together, these technologies support the design of document management applications that combine automated processing with appropriate security and access controls.

An effective document lifecycle management system must combine document extraction, email integration, metadata storage, validation, and expiration monitoring into a coordinated workflow. Once relevant information has been extracted from a document, it can be stored in a structured database to support searching, categorization, and tracking throughout the document's validity period. Automated notifications can then be scheduled according to the identified expiration date. For example, reminders may be configured for 60, 30, 14, and 7 days before expiration, followed by a notification on the expiration date. The reliability of these reminders depends on the accuracy of the extracted dates, the consistency of scheduled processing, and successful notification delivery. A human-in-the-loop approach is therefore important when the extracted information has low confidence, conflicting dates are detected, or the document cannot be interpreted reliably. Documents may be categorized as verified, requiring review, or ignored according to the application's validation rules. Audit records and notification logs can further support traceability by recording document-processing activities and reminder-delivery attempts. Although existing technologies provide capabilities for OCR, multimodal document understanding, email retrieval, authorization, and secure data storage, integrating these capabilities into a complete automated document lifecycle workflow remains an important system-design challenge. DocExpire 2.0: Gmail-Integrated Multimodal AI Document Lifecycle Management and Notification System is designed to address this challenge by combining Gmail attachment processing, AI-based metadata extraction, duplicate detection, human verification, secure token handling, expiration tracking, and automated reminders. The proposed system aims to reduce manual document-management effort and improve the organization, monitoring, and timely renewal of important documents while maintaining appropriate validation and security measures.

CHAPTER 3

METHODOLOGY

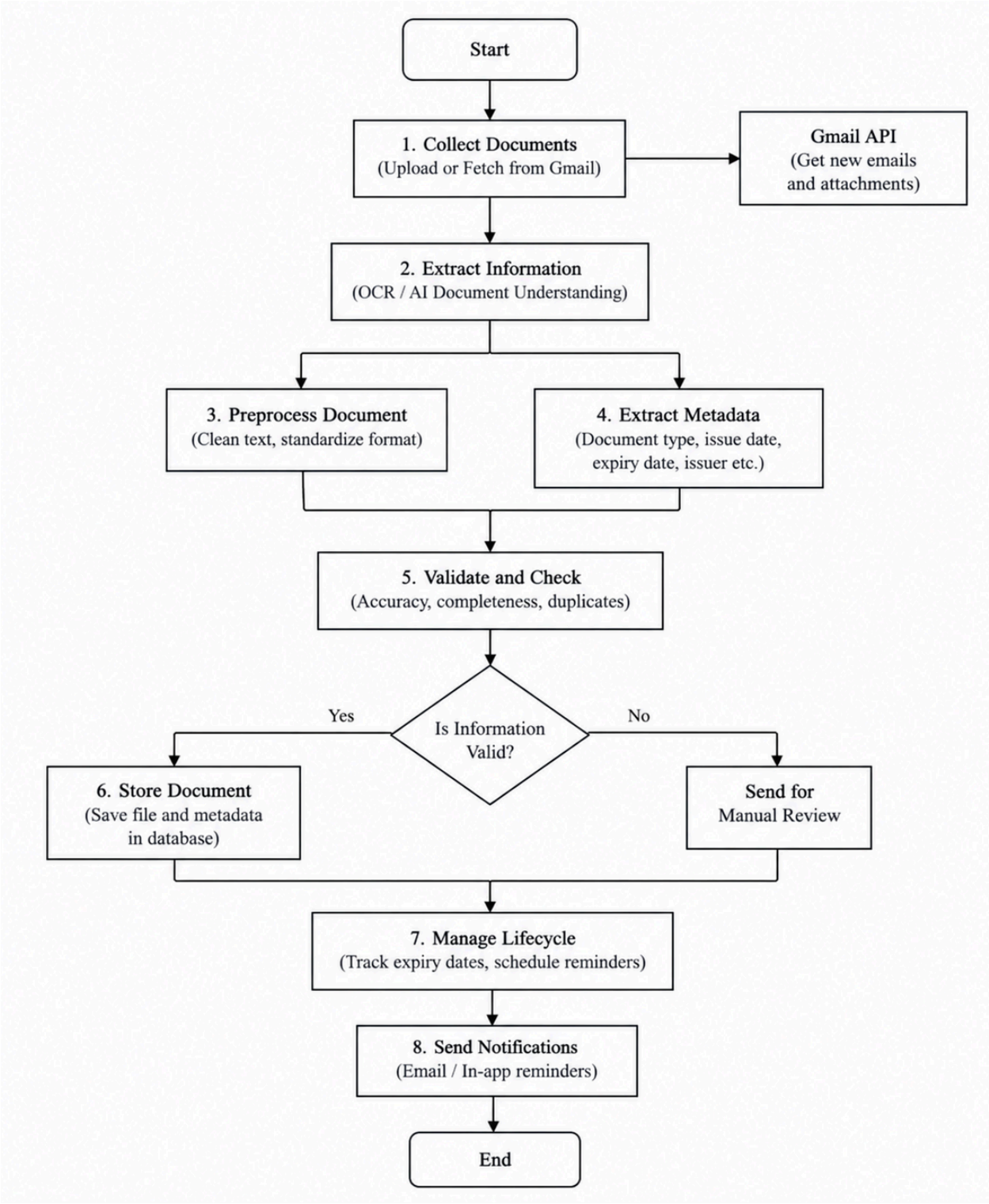
3.1 SYSTEM ARCHITECTURE

The proposed system, Intelligent Campus Energy Demand Forecasting Using Machine Learning, is designed to analyze historical electricity consumption data, identify energy usage patterns, forecast future demand, and provide useful information for campus energy planning. The methodology follows a sequential data-processing and machine-learning workflow in which the output of each stage becomes the input for the next stage. The system begins by collecting electricity consumption readings from CSV files or other suitable meter-data sources. The imported dataset is checked for required columns, valid timestamps, consistent units, duplicate records, missing values, and incorrect readings. After validation, the data is cleaned and arranged in chronological order. Exploratory Data Analysis (EDA) is then performed to understand consumption trends, daily and weekly variations, monthly patterns, and differences between buildings when building information is available. The next stage involves feature engineering, in which useful input variables are created from historical energy readings and timestamps. These features may include the hour of the day, day of the week, month, weekend indicators, previous energy consumption, and rolling averages. Weather conditions, academic calendars, holidays, and occupancy information may also be included when reliable data is available. The prepared features are supplied to a set of forecasting models, including a seasonal-naïve baseline, Linear Regression, Random Forest Regressor, and Gradient Boosting Regressor. An optional Long Short-Term Memory (LSTM) network may be considered when sufficient sequential data and computational resources are available. The candidate models are trained and compared using chronological validation so that future observations do not influence the training process.

The selected model is evaluated using Mean Absolute 13 Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE), where appropriate. Finally, the forecasting output is combined with anomaly identification and optional electricity-cost and carbon-emission estimation modules. An interactive dashboard presents historical consumption, predicted demand, model evaluation results, unusual readings, and sustainability estimates to support decision-making.

Table: System Workflow for Building Energy Demand Forecasting Using Machine Learning

MODULE	**INPUT**	**PROCESS**	**OUTPUT**
Data ingestion	CSV / meter readings	Load file and inspect schema	Raw dataset
Preprocessing	Raw dataset	Validate timestamps, units, duplicates and missing values	Cleaned time-ordered data
Feature engineering	Cleaned data	Derive calendar and lag features without future leakage	Model-ready features
Forecasting	Features and target	Train baseline and candidate models	Predictions
Evaluation	Predictions and actual values	Calculate metrics on chronological holdout	Model comparison
Insights and dashboard	Forecast, history, flags and factors	Visualize demand and estimates	Decision-support views



3.1 System Architecture of the Proposed Energy Demand Forecasting System

3.2 DATASET SELECTION AND PREPARATION

The dataset is the foundation of the proposed forecasting system because the quality and quantity of historical observations directly affect the usefulness of the predictions. The preferred dataset contains timestamped electricity consumption readings for one or more campus buildings, such as classrooms, laboratories, hostels, libraries, and administrative blocks. Each observation should represent electricity consumption during a known time interval or a clearly defined meter reading. The dataset may be obtained from campus energy meters, institutional records, or a suitable publicly available dataset if actual campus readings are unavailable. The final report must identify the actual source used by the project. Before selecting the dataset for model development, its structure and suitability must be examined. Important details include the filename, data source, number of rows and columns, observation period, sampling frequency, target variable, measurement units, missing values, duplicate timestamps, and building coverage. For example, an hourly dataset contains one observation per hour when the records are complete, whereas a daily dataset contains one observation per day. The selected forecasting horizon must match the available data and the project's requirements. If the objective is to forecast the next 24 hours using hourly data, the model must be prepared and evaluated for that horizon rather than evaluated only on a single unspecified future observation. The target variable is the electricity consumption value, preferably expressed in kilowatt-hours (kWh) for the relevant interval. It is essential to determine whether the source contains interval energy consumption or cumulative meter readings. Interval consumption represents the energy used during a particular period, whereas a cumulative meter reading represents the total recorded up to that time. If the dataset contains cumulative readings, interval consumption may be calculated by taking the difference between consecutive readings, provided the meter resets, rollovers, and other irregularities are handled correctly.

3.3 DATA PREPROCESSING

Data preprocessing is the process of converting raw energy records into a consistent and reliable dataset suitable for analysis and forecasting. Raw data may contain missing observations, duplicate entries, inconsistent timestamps, invalid numerical values, incorrect units, or abnormal readings. If these problems are not handled carefully, the forecasting model may learn misleading patterns and produce unreliable predictions. Therefore, preprocessing is carried out before feature engineering and model training. Data loading and structural inspection. The dataset is imported using a suitable data-processing library such as Pandas. The number of rows and columns, column names, data types, sample records, and summary statistics are inspected. The required timestamp and energy-consumption columns are identified, and the numerical target is checked for invalid or non-numeric values. Timestamp conversion and chronological ordering. Timestamp values are converted into a consistent date-time format. Records are then sorted from the earliest to the latest observation. If multiple time zones are present, the timestamps must be converted to a consistent reference. Sorting is important because forecasting depends on the order of observations. Duplicate and missing-value handling. Duplicate records are identified and investigated. Exact duplicate rows may be removed when they are confirmed to be accidental duplicates. Multiple readings at the same timestamp should not be deleted automatically because they may represent separate buildings or meters. Missing timestamps and missing energy values are also identified. If a value is missing, the chosen treatment depends on the sampling frequency, the size of the gap, and the nature of the data. Short gaps may be handled through a documented interpolation or imputation method where appropriate.

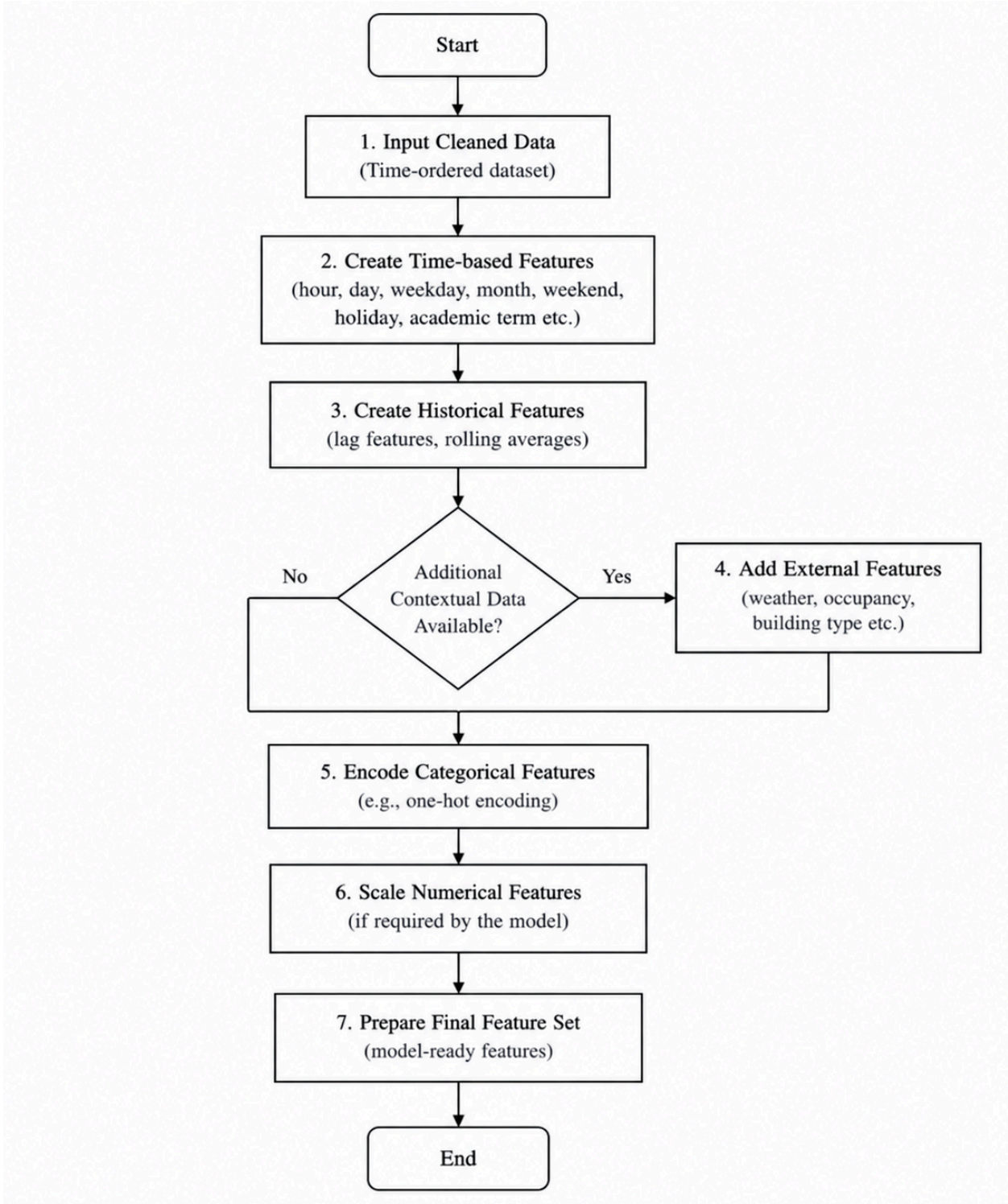
Large gaps may need to remain excluded from certain analyses. Any filling procedure must avoid using information from the future to estimate a past value. 17 Unit consistency and reading validation. Energy readings are checked for consistent units and plausible values. For example, readings reported in watt-hours must be converted appropriately if the target is defined in kilowatt-hours. Negative consumption values, sudden meter resets, and implausibly large values are investigated rather than automatically deleted. The distinction between cumulative meter readings and interval consumption must be preserved. Outlier investigation. An outlier is an observation that differs substantially from the usual pattern. In campus energy data, an unusual peak may result from equipment operation, a special event, or increased occupancy. It may also indicate a faulty reading. Statistical summaries, time-series plots, and domain knowledge are used to investigate such observations. Confirmed data errors are corrected or excluded according to documented rules, while genuine high-consumption events are preserved and may be flagged for later investigation. Data transformation and encoding. Categorical variables, such as building type, may be encoded into numerical representations when the selected model requires numerical inputs. Numerical features may be scaled or standardized when necessary, particularly for algorithms sensitive to feature magnitudes. Tree-based models generally do not require the same scaling as distance-based or gradient-based methods. Any transformation that learns parameters from data must be fitted on the training period and then applied to validation and test periods. Preparation for forecasting. The cleaned observations are arranged in chronological order and checked for gaps or irregular sampling. A forecasting horizon is defined, and the data is prepared for model development. The chronological split is established before fitting data-dependent transformations so that information from validation or test periods does not leak into training.

3.4 EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) is performed to understand how electricity consumption changes over time and to identify patterns that may help forecasting. It uses statistical summaries and visualizations before any final model is selected. This stage also helps identify unusual observations, incomplete periods, and possible relationships between energy use and available contextual variables. The first analysis examines the overall time series by plotting timestamp against energy consumption. This visualization helps identify long-term increases or decreases, recurring peaks, sudden changes, and gaps in the observations. The plot should include clear axis labels, units, and a suitable time scale. A moving average may be added to help reveal broader trends, provided it is used for analysis or constructed without future leakage when used as a model feature. The second analysis examines hourly and daily patterns. If the data is recorded at hourly intervals, average consumption can be compared across the 24 hours of the day. Similarly, weekday and weekend consumption can be compared to investigate differences in campus activity. These comparisons can help identify periods when buildings are normally active or relatively inactive. Monthly summaries can reveal seasonal variation, although meaningful seasonal conclusions require sufficient historical coverage. If building identifiers or building types are available, consumption can also be compared across different buildings. Such comparisons should consider the building's size, function, meter coverage, and observation period where these details are available. Numerical correlation analysis may be performed to explore relationships between consumption and variables such as temperature, occupancy, or time-based features. Correlation alone does not prove that one variable causes changes in energy consumption. The main visualizations recommended for this project are: Time-series line chart: shows historical electricity consumption and changes over time. 19 Hourly consumption chart: compares average demand across different hours of the day.

3.5 FEATURE ENGINEERING

Feature engineering is the process of creating meaningful input variables from existing data so that machine-learning algorithms can learn patterns related to electricity demand. In time-series forecasting, the timestamp and previous energy readings are particularly important because electricity consumption often follows recurring daily and weekly patterns. The features selected depend on the sampling frequency, forecast horizon, data availability, and intended use of the system. Temporal features are derived from the timestamp. These may include hour of day, day of week, day of month, month, weekend indicator, and academic-term or holiday indicator when a verified calendar is available. For example, the hour feature can help a model distinguish morning demand from evening demand. The day-of-week feature can help identify differences between weekdays and weekends. However, these variables should be included only when the timestamp and sampling frequency support them. Lag features represent previously observed energy consumption. For example, if the data is recorded hourly, a lag of one represents consumption in the previous hour, while a lag of 24 represents consumption 24 hours earlier. If the data is recorded daily, a lag of seven represents consumption seven days earlier. Lag features help the model learn from recent usage and potentially recurring patterns. The correct lag values must therefore be selected according to the actual sampling frequency. Rolling statistical features summarize recent consumption over a specified window. A rolling mean of the previous 24 hourly readings, for example, can represent recent average demand. Rolling minimum, maximum, or standard deviation may also be considered when they are useful. These features must be shifted appropriately so that the prediction for a particular time does not include the actual consumption of that time or any later period. Contextual features may include temperature, humidity, occupancy indicators, building type, holidays, and academic schedules.



3.2 Flowchart of Feature Engineering for Energy Demand Forecasting

3.6 FORECASTING MODELS

The forecasting stage uses historical energy consumption and the engineered features to estimate future demand. Multiple models are compared because different algorithms can learn different types of patterns. A simple model may perform well when demand follows regular patterns, while a more complex model may capture nonlinear relationships. The best model cannot be determined in advance; it must be selected through validation using the actual dataset. Seasonal-naïve model: This model is used as a baseline against which more advanced approaches are compared. It predicts future consumption using a value from a comparable previous seasonal period. For example, with hourly data and a daily seasonal period, the forecast for a particular hour may use the consumption recorded at the same hour on the previous day. This approach is computationally inexpensive and provides a reference for determining whether machine learning adds value. Its usefulness depends on whether the chosen seasonal period is meaningful for the dataset. Linear Regression: Linear Regression estimates the relationship between the target consumption and the selected input features. It learns coefficients that minimize prediction errors under the ordinary least-squares formulation. It is simple, relatively easy to interpret, and useful as an initial machine-learning model. However, it may not represent complicated nonlinear relationships between energy use, time, and contextual variables without additional feature transformations. Random Forest Regressor: Random Forest combines the predictions of multiple decision trees trained using randomized samples and feature selection. Individual trees learn decision rules, and their predictions are combined to produce the final estimate. The model can capture nonlinear relationships and interactions between features without requiring the same feature scaling often used by other algorithms.

However, its performance depends on appropriate feature construction and parameter selection, and it may not extrapolate long-term trends effectively. 23 Gradient Boosting Regressor: Gradient boosting constructs a sequence of decision trees, with later trees attempting to reduce errors made by the existing ensemble. This method can capture complex relationships between time-based variables and historical consumption. Parameters such as the number of trees, learning rate, tree depth, and minimum sample requirements can affect its performance. The model should be tuned using the validation period rather than the final test period

Table: Machine Learning Models Used for Building Energy Demand Forecasting and Evaluation

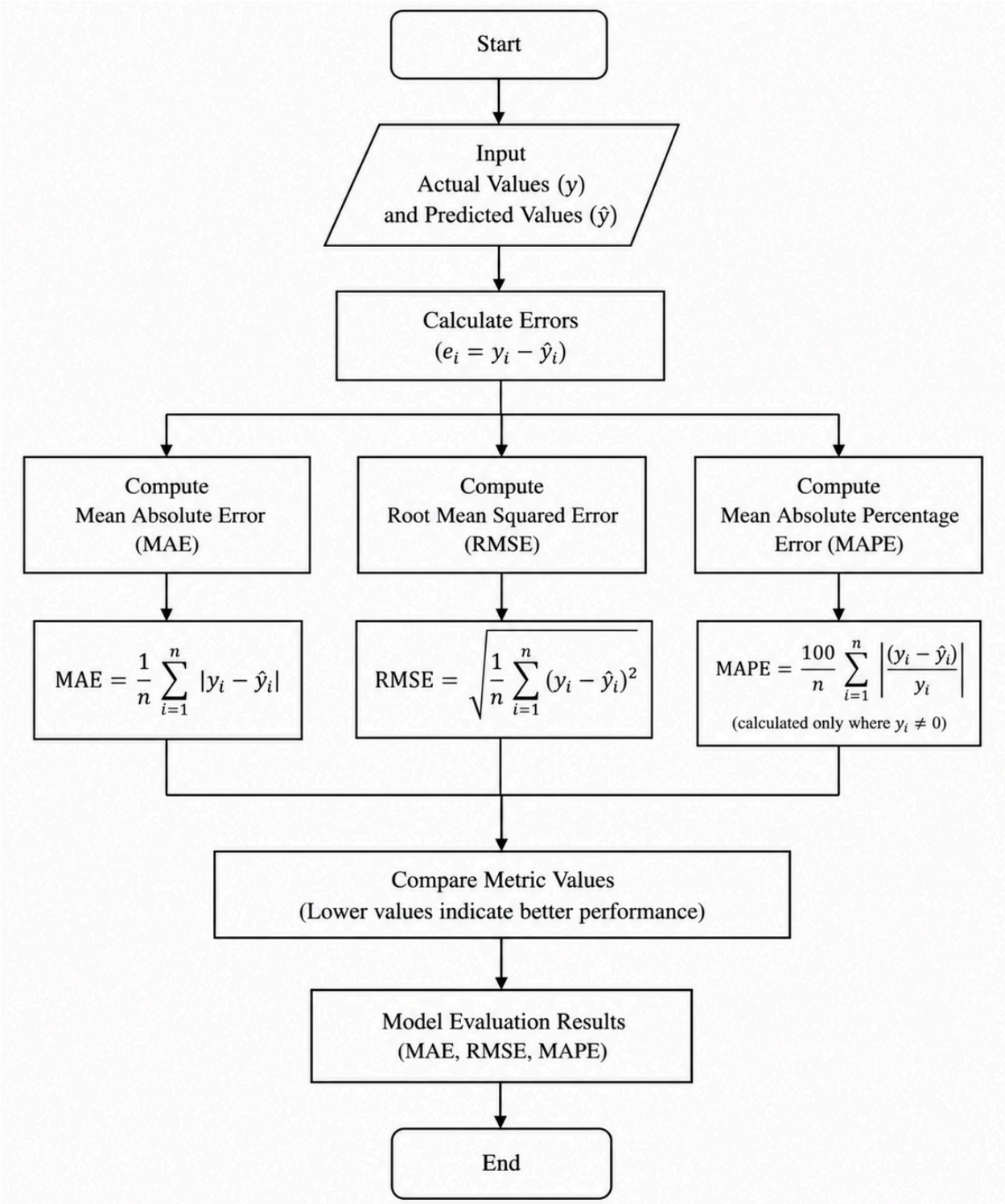
MODEL	**ROLE IN EVALUATION**
Seasonal-naïve	Baseline prediction from a comparable prior period, when the sampling frequency supports it
Linear Regression	Simple, interpretable reference model
Random Forest Regressor	Nonlinear regression using decision-tree ensembles
Gradient-boosting regressor	Sequential tree boosting for complex relationships
Optional LSTM	Consider only if data volume and project resources support it

3.7 CHRONOLOGICAL VALIDATION

Chronological validation is used to assess the forecasting models under conditions that resemble their intended real-world application. In energy forecasting, the goal is to predict future demand from past information. Randomly shuffling observations into training and testing sets can allow later observations to influence training and may produce an unrealistically optimistic estimate of performance. Therefore, the records are arranged by timestamp and separated into consecutive time periods. The earliest portion of the dataset is used for training, the next portion is used for validation, and the latest portion is reserved for final testing. The proportions depend on the dataset's length, frequency, and forecast horizon. A split such as 70% for training, 15% for validation, and 15% for testing can be used as an initial design when sufficient data are available, but it is not a fixed requirement. The actual split dates and observation counts should be documented in the report. The training set is used to learn the model parameters. Any data-dependent preprocessing operations, such as imputation statistics, feature scaling, or encoding rules that learn from observations, are fitted using the training data only. The resulting transformations are then applied to the validation and test periods without refitting them on those periods. This prevents information from later observations from influencing the model during training. The validation set is used to compare candidate models and tune their hyperparameters. For example, different tree depths or learning rates may be compared using validation errors. After model selection, the final model is assessed on the test set, which has not been used to choose the model or its settings. All candidate models must be compared using the same test timestamps and forecasting horizon to make the comparison fair. When the dataset contains enough observations, walk-forward validation may also be performed. In this method, the training window is advanced through time, and predictions are repeatedly evaluated on later observations.

3.8 EVALUATION METRICS

Evaluation metrics are used to measure the difference between actual electricity consumption and the values predicted by the forecasting models. The same evaluation period and forecast horizon must be used for all candidate models so that the comparison is meaningful. Three metrics are proposed: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE), with MAPE used only when its limitations are addressed. Let y represent actual energy consumption for observation , represent the i i $y \hat{} i$ corresponding predicted consumption, and n represent the number of evaluated observations. Mean Absolute Error (MAE) measures the average absolute difference between actual and predicted values. $MAE = 1\ n\ i{=}1\ n \sum |y\ i - y \hat{} i|$ A lower MAE indicates that the predictions are closer to the actual values on average. If the target is measured in kWh, MAE is also expressed in kWh. For example, an MAE of 10 kWh would mean an average absolute prediction error of 10 kWh for the evaluated observations; this is an illustration, not a result from the current project. Root Mean Squared Error (RMSE) calculates the square root of the mean squared prediction error. $RMSE = 1\ n\ i{=}1\ n \sum (y\ i - y \hat{} i)\ 2\ 27$ RMSE gives greater influence to large prediction errors because the errors are squared before averaging. It is useful when large forecasting errors are especially important, such as during periods of high campus demand. RMSE is expressed in the same unit as the target, such as kWh, and a lower value generally indicates better predictions on the same evaluation dataset. Mean Absolute Percentage Error (MAPE) expresses absolute prediction errors as percentages of actual values. $MAPE = 100\ m\ i{\in}S \sum y\ i{-}y \hat{} i\ y\ i$ | | | | | | Here, S is the set of observations with non-zero actual consumption, and m is the number of observations in that set. MAPE can make errors easier to interpret as percentages, but it becomes undefined when actual consumption is zero and may become misleading when actual values are very close to zero. Therefore, the report should state how zero or near-zero values were handled and should not rely on MAPE alone.



3.3 Flowchart of Evaluation Metrics for Forecasting Model Performance

3.9 ANOMALY IDENTIFICATION

Anomaly identification is intended to highlight energy readings that differ substantially from expected consumption patterns. These readings may indicate unusual equipment use, changes in building activity, meter problems, or other events requiring investigation. An anomaly flag does not automatically prove that energy has been wasted or that a fault has occurred. A simple approach is to compare observed consumption with a baseline calculated from comparable historical periods, such as the same hour on previous days or the same time of the week. Another approach is to examine the residual between observed consumption and the forecast produced by the selected model. If the absolute residual exceeds a threshold established using training or validation data, the observation may be flagged for review. A statistical threshold or an appropriate residual distribution can be used, but its definition must be documented and selected without using the final test data to tune it. The anomaly module should record the timestamp, observed consumption, expected consumption, deviation, and reason for flagging. These details can then be displayed on the dashboard to support manual investigation. The threshold should be evaluated against the actual dataset because campus energy use naturally changes with schedules, weather, and operational requirements.

3.10 ELECTRICITY-COST AND CARBON-EMISSION ESTIMATION

The proposed system may estimate electricity costs and associated carbon emissions using the forecasted or observed energy consumption. These calculations provide supplementary decision-support indicators and depend on the availability of valid external factors. They do not, by themselves, demonstrate actual financial savings or reductions in emissions. If E represents electricity consumption in kWh and T represents the applicable electricity tariff in currency per kWh, the simplified energy-charge estimate is: $\textit{Estimated Cost} = E \times T$ For example, if consumption is 100 kWh and the assumed tariff is ₹8 per kWh, the simplified energy-charge estimate is ₹800. This example is illustrative only; it is not a measured result from the project. Actual bills may also include fixed charges, demand charges, taxes, time-of-use pricing, and other adjustments, so the simplified formula must not be presented as a complete bill calculation unless it matches the applicable tariff. If F represents an emissions factor measured in kilograms of carbon dioxide equivalent per kWh, estimated emissions are calculated as: $\textit{Estimated Emissions} = E \times F$ The emissions factor must be obtained from a documented and appropriate source, and the report must specify the relevant region, reporting year, units, and calculation method. Cost and emissions can be estimated separately for historical consumption and future forecasts, but the dashboard must distinguish measured historical consumption from predicted future demand. These estimates can help users explore the potential financial and environmental implications of energy usage without claiming unverified savings. The proposed system may estimate electricity costs and associated carbon emissions using the forecasted or observed energy consumption. These calculations provide supplementary decision-support indicators and depend on the availability of valid external factors.

CHAPTER 4

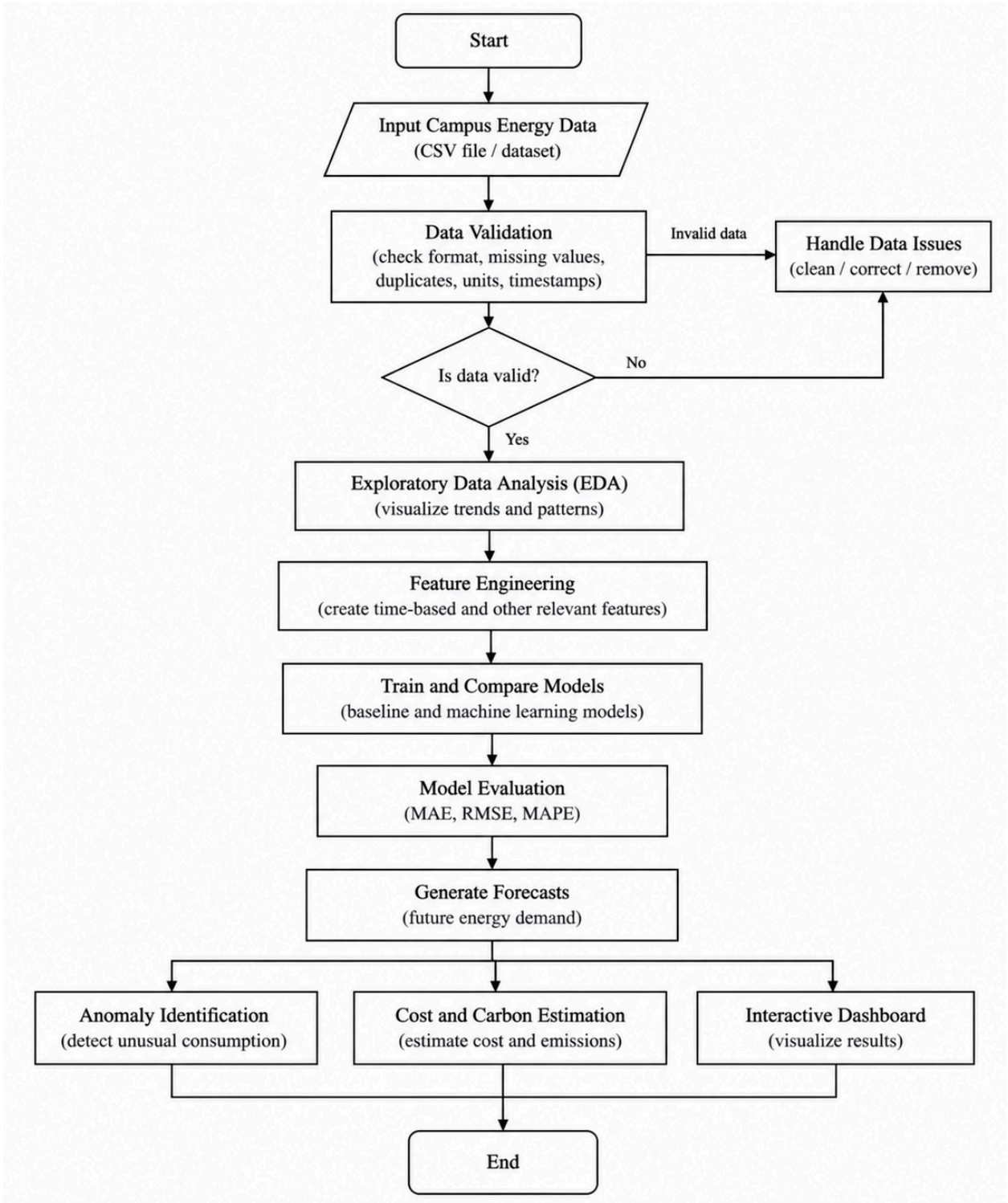
RESULTS AND DISCUSSION

4.1 SYSTEM WORKFLOW

The proposed Intelligent Campus Energy Demand Forecasting Using Machine Learning system follows a structured workflow to process historical electricity consumption data, identify energy usage patterns, forecast future demand, and present the results through an interactive dashboard. The workflow integrates data collection, validation, preprocessing, exploratory data analysis, feature engineering, machine-learning-based forecasting, anomaly identification, and sustainability estimation into a single decision-support system. Each stage performs a specific function, and the output generated by one stage is passed to the next stage to ensure that the forecasting process follows a consistent and organized sequence. The workflow begins with the collection of historical electricity consumption data from a CSV file, energy meter records, or another suitable data source. The dataset provides the historical observations required to understand consumption patterns and train the forecasting models. If additional information, such as building type, weather conditions, occupancy, holidays, or academic schedules, is available and reliable, it can be included to improve the analysis. After importing the dataset, the system examines its structure, column names, data types, timestamps, measurement units, and missing values. Duplicate records, invalid readings, and inconsistent timestamps are also investigated. This validation stage is important because incorrect or incomplete input data can affect both the training process and the reliability of the forecasts. After validation, the data is cleaned and arranged chronologically before exploratory analysis is performed.

Exploratory Data Analysis helps identify changes in consumption over time, recurring daily patterns, weekday and weekend differences, and unusual energy usage. Visualizations and statistical summaries make these patterns easier to 31 understand. The system then creates appropriate features from the available data, including time-based variables and historical consumption values. These features are used as inputs for the machine-learning models. A seasonal-naïve baseline and candidate regression models are trained and evaluated using chronological training and validation periods. Their predictive performance is compared using appropriate error metrics, including Mean Absolute Error and Root Mean Squared Error, together with Mean Absolute Percentage Error when suitable. Following model selection, the selected model is evaluated on a separate test period that was not used for tuning. The resulting forecasts can then be compared with historical consumption to understand expected future demand. An additional anomaly-identification module can flag observations that differ considerably from expected usage. When valid electricity tariff and grid emissions factors are available, the system can also estimate electricity costs and associated carbon emissions. Finally, the dashboard combines these outputs into visual summaries that support energy monitoring, forecasting, and planning. The overall workflow is designed to help users understand historical energy demand and anticipate future requirements. The usefulness of the system depends on the quality of the input data, the forecasting horizon, the performance of the selected model, and the correct interpretation of the results.

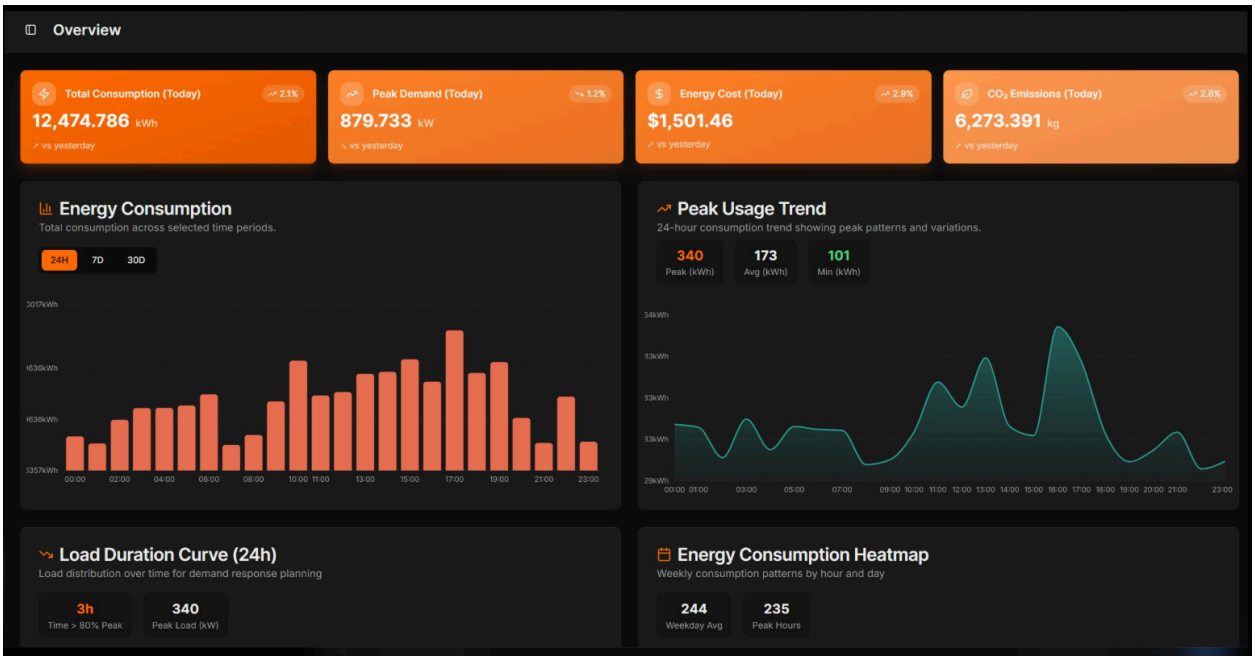
Exploratory Data Analysis helps identify changes in consumption over time, recurring daily patterns, weekday and weekend differences, and unusual energy usage. Visualizations and statistical summaries make these patterns easier to 31 understand. The system then creates appropriate features from the available data, including time-based variables and historical consumption values. These features are used as inputs for the machine-learning models.



4.1 Flowchart of the System Workflow for Intelligent Campus Energy Demand Forecasting Using Machine Learning

4.2 DASHBOARD VIEWS

The proposed dashboard is organized into different views so that users can examine the information generated by the forecasting system according to their needs. Each view serves a specific purpose, while together they provide an overall understanding of campus electricity consumption. The dashboard is designed to connect historical data analysis with future demand estimation and supplementary indicators. The interpretation of each view must be based on the actual data and visualizations produced by the implemented application.



4.3.1 Overview of the AI-Driven Smart Campus Energy Analytics Dashboard

The overview dashboard provides a consolidated summary of the energy consumption information available for the selected observation period. It is intended to help users understand the general status of the analyzed dataset without requiring them to examine individual records.

Depending on the fields and calculations supported by the implementation, the overview may display total energy consumption, average consumption, peak recorded demand or consumption, forecasted energy usage, and estimated cost or emissions. The exact indicators should reflect the actual calculations performed by the system.

The overview view provides an entry point for examining energy-related information and identifying areas that may require further investigation. For example, a user may select a particular observation period and examine the corresponding consumption indicators before moving to the detailed trend or model-comparison views. If building-level information is available, the dashboard may also allow users to filter the results by building. Such filtering can help users compare different areas of the campus, provided the underlying data supports a meaningful comparison.

The overview should clearly distinguish measured values from predicted values and calculated estimates. Historical energy consumption is derived from recorded data, whereas forecasted consumption is generated by a model. Similarly, cost and emissions values are estimates based on the chosen tariff and emissions factor. Maintaining this distinction is important because users should be able to identify which values are directly observed and which depend on model predictions or external assumptions.

The screenshot should be discussed according to the indicators, charts, filters, and labels that are actually visible in the implemented dashboard. If the overview includes summary cards or visual comparisons, explain what each one represents and how it helps users understand the selected energy data. Do not describe indicators that are not present in the screenshot or claim measured improvements that have not been verified.

4.2.1 Daily Campus Energy Consumption Trend

The daily campus energy consumption trend view is intended to show how electricity consumption changes across successive days in the selected observation period. A time-series visualization places the date on the horizontal axis and energy consumption on the vertical axis, using the appropriate unit for the data. This representation allows users to examine variations in daily consumption and identify periods with relatively high or low usage. It can also help reveal missing observations, abrupt changes, and possible recurring patterns in the historical readings.

When the dashboard includes forecast values, the predicted consumption should be displayed separately from historical observations. Clear legends and labels help users understand the point at which recorded data ends and the forecast begins. Comparing the predicted trend with the historical pattern may help users examine whether the forecast reflects recurring changes in demand. However, the apparent similarity between a forecast and historical data does not independently establish its accuracy. Actual forecasting performance must be measured by comparing predictions with the corresponding actual values over an appropriate held-out period.The daily trend view can support energy planning by making changes in consumption easier to identify. For example, a period of unusually high usage can be examined alongside building activities, academic schedules, or other available information.A low-consumption period may similarly reflect reduced campus activity or incomplete readings. The chart should therefore be interpreted in the context of data quality and operational conditions rather than assuming that every peak indicates waste or every decrease represents an improvement.The anomaly identification view is intended to display energy readings that differ from the expected consumption pattern. It may present a table or chart containing the timestamp, observed consumption, expected consumption, deviation, and reason for flagging. The purpose of this view is to make unusual observations easier to inspect and compare with the surrounding historical data.

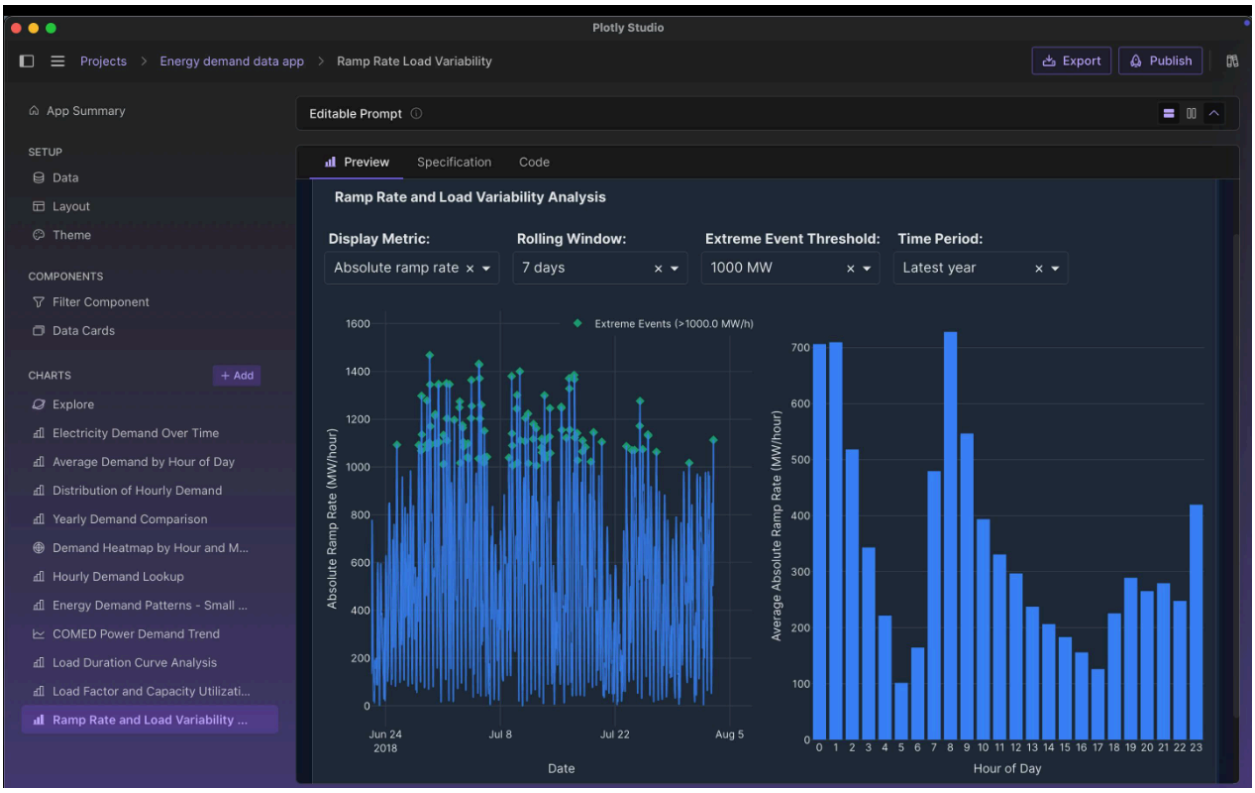


Figure 4.2 – Daily Campus Energy Consumption Trend

The accompanying discussion should identify the time period shown, the unit used on the vertical axis, whether the displayed values represent actual consumption or predictions, and any visible patterns supported by the chart. If the chart contains only historical data, it should be described as a historical consumption trend rather than a forecasting result.

The anomaly identification view is intended to display energy readings that differ from the expected consumption pattern. It may present a table or chart containing the timestamp, observed consumption, expected consumption, deviation, and reason for flagging. The purpose of this view is to make unusual observations easier to inspect and compare with the surrounding historical data.

4.2.2 Forecast and Model-Comparison View

The forecast and model-comparison view is intended to present the future demand estimates and the evaluation results of the candidate forecasting models. The view may compare the seasonal-naïve baseline, Linear Regression, Random Forest Regressor, and Gradient Boosting Regressor using MAE, RMSE, and appropriate MAPE values. Displaying the metrics together helps users understand how the models perform under the same evaluation conditions. For the comparison to be meaningful, each model must be evaluated using the same test timestamps and forecast horizon.

A model with a lower MAE produces smaller absolute prediction errors on average, while a lower RMSE indicates a lower root mean squared error and generally fewer or smaller large deviations. MAPE can provide a percentage-based comparison when the actual values are suitable for its calculation. The metrics should be interpreted together because a model may perform differently depending on whether average errors or larger errors are emphasized. A more complex model should not automatically be considered superior to a simpler model unless the evaluation results support that conclusion.

The forecast visualization should distinguish the historical observations from future predictions and identify the forecast period. If actual values are available for a previously held-out period, predicted and actual values can be plotted together to examine their differences. The report should include the measured model metrics and describe the selected model only after the actual results have been obtained. The forecast and model-comparison view is intended to present the future demand estimates and the evaluation results of the candidate forecasting models. The view may compare the seasonal-naïve baseline, Linear Regression, Random Forest Regressor, and Gradient Boosting Regressor

4.2.3 Anomaly Identification View

The anomaly identification view is intended to display energy readings that differ from the expected consumption pattern. It may present a table or chart containing the timestamp, observed consumption, expected consumption, deviation, and reason for flagging. The purpose of this view is to make unusual observations easier to inspect and compare with the surrounding historical data.

The anomaly results should be interpreted carefully because unusual consumption does not necessarily indicate energy wastage or an equipment fault. Changes in building schedules, special events, equipment use, and meter problems may all contribute to unusual readings. The flagged observations therefore act as indicators for further investigation. If the implementation uses a threshold based on historical consumption or forecast residuals, the report should explain how that threshold was selected and how its suitability was evaluated.

The anomaly view may help campus administrators or energy-management personnel identify periods that deserve attention. However, the effectiveness of the module must be established through actual testing, and any claimed detection performance should be supported by verified observations or suitable evaluation data.

The anomaly identification view is intended to display energy readings that differ from the expected consumption pattern.

4.2.4 Sustainability and Energy-Cost View

The sustainability view is intended to summarize the estimated financial and environmental implications of electricity consumption. When an applicable tariff is available, the system can calculate a simplified electricity-charge estimate from energy consumption. When a suitable grid emissions factor is available, it can calculate estimated carbon emissions. The dashboard may display these values alongside the selected observation period or future forecast, allowing users to examine the implications of different levels of energy demand.

The interpretation of this view depends on the accuracy of the input factors. The tariff must correspond to the applicable electricity pricing arrangement, and the emissions factor must be appropriate to the relevant region and reporting period. The dashboard should identify the source and units of these factors so that users can understand the assumptions behind the estimates. If the system uses a simplified tariff calculation, it must not imply that the result represents the complete electricity bill unless all relevant charges are included.

The sustainability view can support planning by showing how projected energy use relates to estimated expenditure and emissions. Nevertheless, forecasted reductions in demand should not be presented as achieved savings. Actual cost savings and emissions reductions can be claimed only after they have been measured and compared against a suitable baseline. In the current project report, these calculations should be described as proposed estimates unless actual outputs and their input factors have been verified.

4.3 COST AND CARBON ESTIMATION

Cost and carbon estimation provides supplementary analysis by translating electricity consumption into estimated financial expenditure and associated greenhouse-gas emissions. The purpose of these calculations is to help users understand the possible implications of campus energy demand in addition to the energy values themselves. These calculations may be applied to historical consumption or predicted future consumption, but the results must be clearly labelled according to the type of energy value used.

Electricity-cost estimation uses the energy consumed during the relevant period and the applicable tariff per unit of electricity. Let E represent energy consumption measured in kilowatt-hours (kWh), and let T represent the tariff expressed in currency per kilowatt-hour. The simplified energy-charge estimate is calculated as follows:

$$EstimatedCost = E \times T$$

For example, if a hypothetical building consumes 500 kWh and the assumed tariff is ₹8 per kWh, the estimated energy charge is ₹4,000. This example illustrates the calculation only and does not represent the actual consumption, tariff, or cost of the project campus. In a real implementation, the tariff must be obtained from a documented and applicable source. If the electricity pricing structure includes fixed charges, demand charges, taxes, time-of-use rates, or other adjustments, these must be considered when estimating the total bill. Otherwise, the output should be described specifically as a simplified energy-charge estimate.

Carbon-emission estimation calculates the emissions associated with electricity consumption using an appropriate grid emissions factor.

Let E represent electricity consumption in kWh and F represent the emissions factor in kilograms of carbon dioxide equivalent per kWh. The estimated emissions are calculated using the following formula:

$$EstimatedEmissions = E \times F$$

For example, if hypothetical electricity consumption is 500 kWh and an illustrative emissions factor is 0.7 kg CO_2e/kWh, the estimated emissions are 350 kg CO_2e. The factor used in this example is for explaining the formula only and must not be treated as the project's verified emissions factor. The actual factor must be selected from a documented source that is suitable for the relevant electricity grid, geographical region, and reporting year.

Both calculations depend on accurate energy readings and appropriate external inputs. If historical consumption is used, the result represents an estimate associated with the recorded energy usage. If predicted consumption is used, the result represents a projection based on the forecasting model and the assumed tariff or emissions factor. These two types of results should not be mixed because a forecast is uncertain and may differ from the actual energy demand. The dashboard should therefore display the calculation period, input factor, unit, and whether the value is historical or predicted.

The estimated cost and emissions may help users explore the potential implications of future electricity demand and compare different consumption scenarios. However, these calculations alone cannot prove that the forecasting system has reduced electricity consumption, lowered expenditure, or reduced carbon emissions. Such outcomes require an actual implementation, a defined comparison baseline, and measured evidence over an appropriate period. The proposed module should therefore be understood as a decision-support feature that complements the forecasting system rather than a direct measurement of achieved sustainability improvements.

Table: Energy Consumption, Electricity Tariff, and Emissions Factor Data Requirements

INPUT	UNIT	REQUIRED DOCUMENTATION
Energy consumed (E)	kWh	Meter interval and reporting period
Electricity tariff (T)	Currency/kWh	Applicable tariff, region, effective date and whether charges are simplified
Emissions factor (F)	kg CO_2e/kWh	Source, region, reporting year and methodology

4.5 TOOLS AND TECHNOLOGIES

Table: Technologies and Tools Used for Building Energy Demand Forecasting Using Machine Learning

TECHNOLOGY	PROPOSED USE
Python 3	Main programming language
Pandas and NumPy	Data loading, cleaning and numerical operations
Matplotlib / Seaborn / Plotly	EDA and visualizations
Scikit-learn	Preprocessing, regression and evaluation
TensorFlow / Keras	Optional sequence-model implementation
Streamlit or Power BI	Interactive dashboard
SQL / PostgreSQL	Optional structured storage
Jupyter / VS Code / GitHub	Experimentation, development and version control

These are candidate technologies in the project plan. The final report should state which tools were actually used; unused tools should remain labelled as proposed or be removed.

CHAPTER 5

CONCLUSIONS AND SUGGESTIONS FOR FUTURE WORK

5.1 CONCLUSION

The project titled Intelligent Campus Energy Demand Forecasting Using Machine Learning focuses on developing a data-driven approach to analyzing campus electricity consumption and predicting future energy demand. Electricity is required for the daily operation of classrooms, laboratories, libraries, hostels, administrative buildings, and other campus facilities. Variations in building usage, operating schedules, weather conditions, and academic activities can influence electricity consumption over time. Understanding these variations is important for effective energy planning and resource management. The proposed system aims to address the challenges associated with manually analyzing historical energy readings by integrating data preprocessing, exploratory data analysis, machine-learning-based forecasting, anomaly identification, electricity-cost estimation, carbon-emission estimation, and interactive visualization into a unified workflow. The methodology begins with collecting historical electricity consumption data and validating its quality before using it for analysis. Data preprocessing is intended to handle missing values, duplicate records, inconsistent timestamps, and invalid readings according to documented rules. Exploratory Data Analysis helps examine consumption trends and identify temporal patterns that may be useful for forecasting. Feature engineering transforms the cleaned data into meaningful inputs, including time-based variables and historical consumption features. These inputs are then used to develop and compare candidate forecasting models. The proposed models include a seasonal-naïve baseline, Linear Regression, Random Forest Regressor, and Gradient Boosting Regressor, with LSTM considered as an optional extension when sufficient sequential data and computational resources are available. 43 An important aspect of the proposed methodology is chronological validation, which preserves the order of observations and reduces the risk of future information influencing model

training. The forecasting models are intended to be compared using Mean Absolute Error, Root Mean Squared Error, and Mean Absolute Percentage Error where appropriate. This evaluation process provides a systematic way to identify a suitable model based on measured predictive performance rather than assuming that a particular algorithm is always superior. The forecasting output can then be presented through an interactive dashboard, allowing users to examine historical consumption, predicted demand, and model evaluation results in a more understandable format. The proposed system also includes anomaly identification and sustainability-estimation features. Anomaly identification is intended to highlight consumption readings that differ from expected usage patterns so that they can be investigated further. Electricity-cost and carbon-emission estimation can provide additional information about the possible financial and environmental implications of energy consumption when valid tariff and emissions-factor inputs are available. These features are intended to support decision-making by presenting energy information in a structured and accessible manner. However, unusual readings must not automatically be treated as equipment faults or energy wastage, and calculated cost and emissions values must be interpreted accordingly.The forecasting models are intended to be compared using Mean Absolute Error, Root Mean Squared Error, and Mean Absolute Percentage Error where appropriate. This evaluation process provides a systematic way to identify a suitable model based on measured predictive performance rather than assuming that a particular algorithm is always superior. The forecasting output can then be presented through an interactive dashboard, allowing users to examine historical consumption, predicted demand, and model evaluation results in a more understandable format.

5.2 SUGGESTIONS FOR FUTURE WORK

The proposed system can be improved in the future by integrating real-time electricity consumption data from campus smart meters. Automated collection would reduce the need for repeated manual dataset uploads and allow the application to process new readings at regular intervals. A database-backed architecture could be introduced to store historical observations, forecasts, model evaluation results, and anomaly records. This would make it easier to maintain a continuous record of energy usage, compare consumption across different periods, and update predictions as new data becomes available. The automated pipeline should also include validation and error-handling mechanisms to identify incomplete readings, irregular timestamps, and data collection failures. Another important direction for future development is the inclusion of reliable contextual information. Weather conditions, academic calendars, holidays, occupancy indicators, and building characteristics may help explain changes in electricity demand. Integrating these variables could allow the forecasting models to represent campus operating conditions more effectively. However, each additional feature should be evaluated to determine whether it improves forecasting performance and whether its data is reliable and available at the time of prediction. Future versions could also investigate separate models for individual buildings because classrooms, laboratories, hostels, and administrative blocks may have different consumption patterns and operating schedules. The forecasting component can be extended by evaluating additional machine-learning and time-series algorithms. The current approach provides a foundation for comparing a seasonal-naïve baseline with regression-based models and, where appropriate, sequence-learning methods. Future experiments could use walk-forward validation to examine how model performance changes across different time periods. Separate forecasting models could also be developed for average consumption and peak demand, as both are relevant to energy planning.

The selection of a final model should continue to 46 depend on measured test performance, reliability, interpretability, and practical requirements rather than model complexity alone. Future work could improve anomaly identification by developing building-specific consumption baselines and refining the thresholds used to flag unusual readings. Historical consumption patterns could be compared with current observations to identify deviations that require further examination. Additional contextual information could help distinguish legitimate increases in energy demand from possible measurement problems or unusual operating conditions. The system could also be extended with configurable alerts when observed or predicted consumption exceeds a documented threshold. Such alerts should be treated as prompts for investigation and should not automatically classify a reading as a fault or evidence of energy wastage. The electricity-cost and carbon-emission estimation modules could be enhanced by integrating verified electricity tariffs and appropriate official emissions factors. The cost-estimation method could be extended to account for applicable fixed charges, demand charges, time-of-use pricing, and other billing components where the necessary information is available. The emissions-estimation process could be updated using factors appropriate to the relevant electricity grid and reporting period. Another important direction for future development is the inclusion of reliable contextual information. Weather conditions, academic calendars, holidays, occupancy indicators, and building characteristics may help explain changes in electricity demand. Integrating these variables could allow the forecasting models to represent campus operating conditions more effectively. However, each additional feature should be evaluated to determine whether it improves forecasting performance and whether its data is reliable and available at the time of prediction.

REFERENCES

[1] T. Lodderstedt et al., “Best Current Practice for OAuth 2.0 Security,” RFC 9700, IETF, 2025. https://www.rfc-editor.org/rfc/rfc9700.html

[2] N. Sakimura et al., “Proof Key for Code Exchange by OAuth Public Clients,” RFC 7636, IETF, 2015. https://www.rfc-editor.org/rfc/rfc7636.html

[3] D. Hardt, “The OAuth 2.0 Authorization Framework,” RFC 6749, IETF, 2012. https://www.rfc-editor.org/rfc/rfc6749.html

[4] M. Dworkin, “Recommendation for Block Cipher Modes of Operation: GCM and GMAC,” NIST SP 800-38D, 2007. https://csrc.nist.gov/pubs/sp/800/38/d/final

[5] Tesseract OCR, “Tesseract OCR Documentation.” https://tesseract-ocr.github.io/

[6] Google AI for Developers, “Document Understanding,” Gemini API Documentation. https://ai.google.dev/gemini-api/docs/document-processing

[7] Google for Developers, “Get Message Attachments,” Gmail API Documentation. https://developers.google.com/workspace/gmail/api/reference/rest/v1/users.messages.attachments/get

[8] Google for Developers, “List Mailbox History,” Gmail API Documentation. https://developers.google.com/workspace/gmail/api/reference/rest/v1/users.history/list

WORK AREA	THATHIKA (7376241CS442)	VARSHNIE (7376241CS451)
Problem definition and planning	Review problem statement, objectives and project scope	Review requirements, deliverables and project plan
Dataset preparation and EDA	Dataset inspection, cleaning and visualization (confirm actual work)	Data quality checks, feature planning and visualization (confirm actual work)
Model development	Baseline/model implementation and metric calculation (confirm actual work)	Candidate model comparison and evaluation (confirm actual work)
Dashboard and insights	Forecast and EDA dashboard views (confirm actual work)	Anomaly and sustainability views (confirm actual work)
Testing and documentation	Record test cases, data assumptions and results	Consolidate evaluation, limitations, references and final report

CHAPTER 8 – ORIGINALITY SCORE

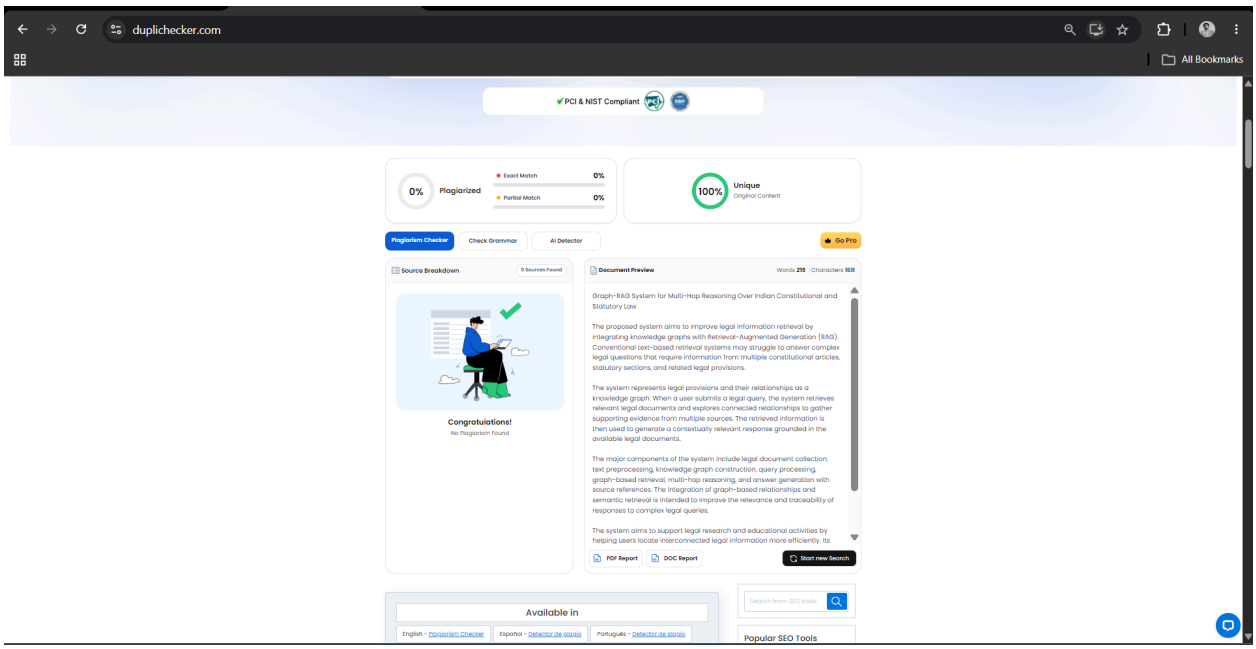


FIGURE 8.1 – ORIGINALITY REPORT

ITEM	VALUE TO COMPLETE FROM ACTUAL REPORT
Checker	duplichecker.com
Originality score	100
Date generated	13.10.2026

APPENDIX A – ENERGY FORECASTING QUESTION SET

Use these as starting points only after confirming that the relevant provisions are present in the actual corpus. For each question, manually identify the expected evidence provisions before running the system.

1. Which constitutional provision establishes the relevant right, and what related provision defines or limits its application?
2. Which provision defines a term used by another provision, and how does that definition affect interpretation of the second provision?
3. Does a statutory section refer to another section for procedure, exceptions, or definitions? Identify both provisions and explain the connection.
4. What evidence is available in the indexed corpus for this question, and which parts remain unsupported?
5. Does the answer depend on a specific amendment or effective date? Identify the version represented in the corpus.

APPENDIX B – USER GUIDE

Open the locally deployed or hosted prototype using the actual URL or launch procedure.

1. Enter a specific legal question and include the relevant Act or provision name when known.
2. Submit the question and wait for the retrieval and answer-generation process to complete.
3. Inspect the answer, provision identifiers, source excerpts, and any uncertainty message.
4. Open each cited source and verify that the cited wording supports the claim.
5. Record unsuccessful queries, missing provisions, incorrect links, and response time for evaluation.